SRM UNIVERSITY AP
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**PROJECT TITLE**

# SMART VENDING MACHINE


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# ABSTRACT

This project presents an IoT-based Smart Object Dispenser that automatically dispenses selected objects after a digital payment is confirmed. The system integrates a web-based front-end, Firebase Realtime Database, and an ESP32 microcontroller for automation.

Users access the web application to view available items (Melody, Kopiko, Alpenliebe), add products to their cart, and make payments through UPI (Google Pay / Phone Pe / Paytm). The transaction details — including the product name, amount, payment status, and timestamp — are stored in Firebase.

An ESP32 continuously monitors the database and detects when the payment status changes to “Completed”. It then activates the corresponding servo and motor mechanism to dispense the correct object. This system demonstrates an affordable, cashless, real-time vending/dispenser model suitable for educational and commercial environments.

# INTRODUCTION:

In recent years, the integration of Internet of Things (IoT) technologies with automation and cashless payment systems has revolutionized the way products and services are delivered.

Traditional vending machines, though effective, are expensive, mechanically complex, and rely on coins, cards, or proprietary payment systems. These limitations make them unsuitable for small businesses, institutions, and rural areas where cost efficiency and simplicity are essential.

To overcome these issues, this project introduces an IoT-Based Smart Object Dispenser, a compact, affordable, and cloud-connected system that dispenses objects automatically after a QR/UPI payment is completed.

The system leverages Firebase Realtime Database as a communication bridge between the web application and the ESP32 microcontroller, ensuring real-time synchronization of user actions and hardware response.

The Smart Object Dispenser allows users to:

View available products (like snacks or beverages) on a web interface.

Add desired items to a shopping cart and view the total cost.

Make payment securely through a UPI QR code using mobile applications such as Google Pay, PhonePe, or Paytm.

Confirm payment by clicking “I Have Paid,” which triggers real-time updates in Firebase.

Automatically receive the selected item through servo- and motor-based dispensing mechanisms controlled by the ESP32.

# METHODOLOGY

## 1. System Overview

1.**Web Front-End**

HTML webpage hosted on Firebase.
Displays products dynamically from Firebase.
Handles cart management and UPI QR generation.

2.**Cloud Backend (Firebase)**

**Stores:**
Product data (machines/M001/products)
Base UPI link (machines/M001/qr)
Payment records (payments/latest)
Synchronizes data between website and ESP32.

3.**Embedded Hardware (ESP32 Controller)**

Monitors Firebase in real-time.
Activates servos and motors to dispense selected items.
Updates the database with the new payment status (Dispensed).

## 2. Hardware Components

| Component | Function |
|---|---|
| ESP32 | Main IoT controller with Wi-Fi capability |
| SG90 Servo Motors (x3) | Dispense individual items |
| L298N Motor Driver | Controls DC motor/conveyor |

| | |
|---|---|
| DC Motor | Pushes item from slot |
| Relay Module | Controls motor/servo power |
| 5V Power Supply | Provides power to motors and ESP32 |

**3.Connections:**

| **Component** | **ESP32 Pin** |
|---|---|
| Servo 1 (Coke) | GPIO 13 |
| Servo 2 (Lays) | GPIO 12 |
| Servo 3 (Pepsi) | GPIO 14 |
| L298N IN1 | GPIO 25 |
| L298N IN2 | GPIO 26 |
| Relay IN | GPIO 33 |

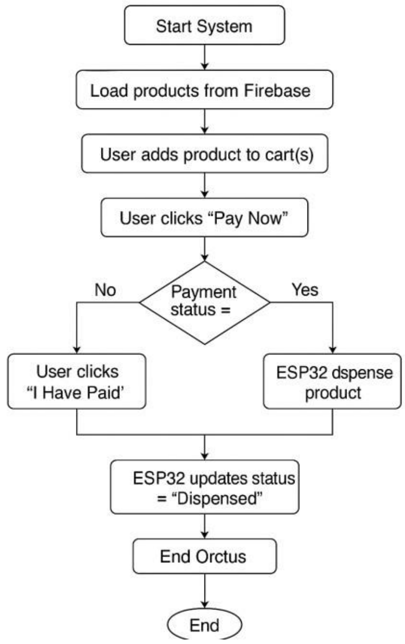
## 3. Front-End Operation

- Display Products: Dynamically loaded from Firebase.
- Cart System: Users can add/remove items and view the total.
- Pay Now: Generates UPI link using stored UPI ID and total amount.
- I Have Paid: Updates Firebase status to Completed.

## 4. ESP32 Function

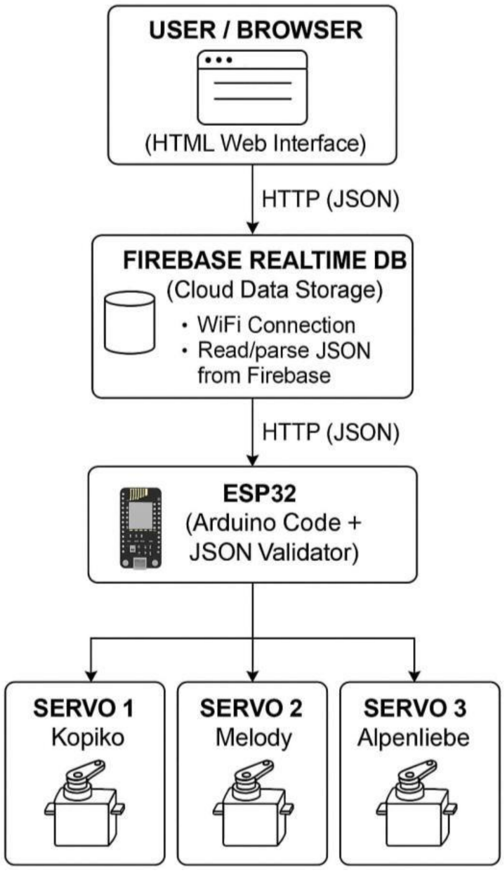
- Connects to Wi-Fi and Firebase REST API.
- Continuously monitors /payments/latest/status.
- When "Completed":
    - Reads the product name.
    - Activates the corresponding servo and DC motor.
    - Updates status → "Dispensed"

# FLOW CHART:




The flowchart illustrates the complete working of the Smart Object Dispenser system. The process begins when the system is powered on, initializing the ESP32, Firebase connection, and all hardware components. The website then loads available products such as Coke, Lays, and Pepsi from the Firebase database, displaying their names, prices, and images. The user selects desired products and adds them to the cart, after which the total amount is calculated automatically. When the user clicks "Pay Now," a UPI payment QR code or link is generated, and a payment record with status "Pending" and timestamp is stored in Firebase. Once the payment is completed, the user clicks "I Have Paid," changing the payment status in Firebase to "Completed." The ESP32 continuously monitors Firebase for payment updates and, upon detecting a "Completed" status, activates the respective servo motor to dispense the selected product into the output tray. After successful dispensing, ESP32 updates the payment status to "Dispensed," completing the process and resetting the system for the next user.

# ARCHITECTURE




### 1. User / Browser (HTML Web Interface)

- The user opens a webpage with buttons for Kopiko, Melody, and Alpenliebe.
- When the user selects an item, the HTML page sends a JSON command to Firebase Realtime Database through an HTTP request.

### 2. Firebase Realtime Database (Cloud Layer)

- Acts as the central cloud server for data exchange.
- It stores the value of "status" (e.g., "Kopiko").
- This database is updated instantly and can be accessed from anywhere.

### 3. ESP32 (Arduino Code + JSON Validator)

- The ESP32 connects to Firebase through Wi-Fi.
- It sends an HTTP GET request to read the JSON data from Firebase.
- Using the ArduinoJson library (JSON Validator), it checks and extracts the "status" value.

- Based on the product name received, it activates the corresponding servo motor.

**4. Servo Motors (Actuator Layer)**

- Each servo motor is linked to one candy slot:
  Servo 1 → Kopiko
  Servo 2 → Melody
  Servo 3 → Alpenliebe
- The selected servo rotates to dispense the chosen item.
- After dispensing, the ESP32 updates Firebase with "status": "Completed".

# RESULTS :

**1. Web Interface**

- Products displayed correctly from Firebase.
- Cart updates dynamically as items are added/removed.
- UPI payment link generated instantly with the correct amount.
- Real-time timestamp logged in Firebase.

Fig-1: This figure represents the web page of the object dispenser

## 2. Firebase Logs

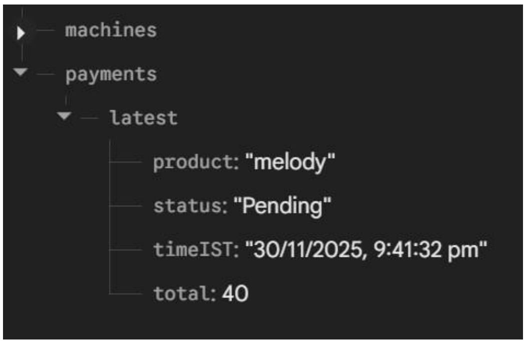



Fig-2: This figure represents the before payment in the firebase

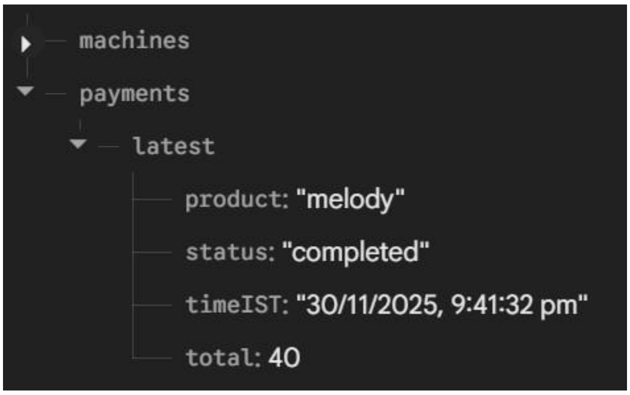



Fig-3: This figure represents the after the completion of payment in the firebase

### 3. Hardware Action

- When payment is confirmed:
  - The ESP32 activates the corresponding servo.
  - DC motor pushes the item forward.
  - Relay switches power on/off safely.

```
🚀 ESP32 Smart Vending firmware started
Raw status JSON: "Completed"
Current status: Completed
Product to dispense: Melody
🚚 Dispensing: Melody
PATCH response: {"status":"Dispensed"}
✅ Updated status to Dispensed
Raw status JSON: "Dispensed"
Current status: Dispensed
```

Fig-4: This figure represents after completion of dispence the item

# CONCLUSION :

The **IoT-Based Smart Object Dispenser** successfully integrates cloud technology, online payment, and automated dispensing in one system. It provides a **cashless, realtime, and efficient** vending solution suitable for small-scale applications.

The project demonstrates:

- Real-time IoT control using Firebase.
- QR/UPI-based payment integration.
- Reliable dispensing mechanism with ESP32.

This system can be extended with features like **stock monitoring**, **multiple vending slots**, or **payment verification through APIs** for deployment in public spaces, schools, or offices.